Causal Discovery: Foundations and Algorithms

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Session 1: Fundamentals of Causality

Learning Outcomes

By the end of this session, you will be able to:

- **Define** causality in terms of structural mechanisms.
- **Identify** the three fundamental structures: Chains, Forks, and Colliders.
- **Understand** the "Conditioning Intuition": which structures allow information flow and which block it.
- **Differentiate** between RCTs, Quasi-experiments, and Causal Discovery.

What is Causality?

- **Operational definition (intervention):** X causes Y if actively setting X changes the distribution of Y:

$$P(Y \mid do(X = x_1)) \neq P(Y \mid do(X = x_2)).$$

- **Asymmetry:** causality is directional ($X \to Y$ is not the same as $Y \to X$).
- **Structural (SCM) perspective:** causality is about **mechanisms**—structural equations that remain **invariant** under many changes in context.
- **Interventions replace mechanisms:** $do(X{=}x)$ modifies the system by setting X externally while leaving other mechanisms unchanged.

Examples of Causality

- **Physical:** Turning the steering wheel (X) causes the car to turn (Y).
- **Biological:** A virus (X) causes an immune response (Y).
- **Policy:** Increasing the minimum wage (X) may cause changes in employment rates (Y).

The Intervention Test
If we can change Y by manipulating X, then X is a cause of Y.

Correlation is not (necessarily) Causation

Two variables can be statistically related without a direct causal link:

1. **Spurious Correlation:** Ice cream sales and drowning (both caused by Heat).
2. **Reverse Causation:** Firefighters and fire damage.
3. **Coincidence:** Statistically significant but meaningless links in high-dimensional data.

Statistics describes "What Is." Causality explains "Why."

Key Causal Roles (The "Big Three")

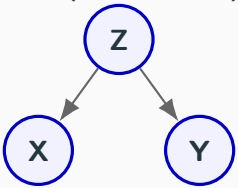
Every complex DAG is built from these three basic structures:

Chain (Mediator)

Path: Open

X affects Y through M

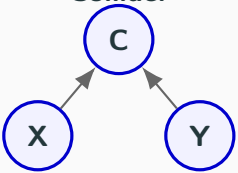
Fork (Confounder)



Path: Open

Z causes both X and Y

Collider



Path: **Closed**

X and Y share a common effect.

Examples of Key Causal Roles

- **Mediator:** Education $\rightarrow$ Skillset $\rightarrow$ Salary.
- **Confounder (Fork):** Genetics $\rightarrow$ Smoking *and* Genetics $\rightarrow$ Lung Cancer. (A classic argument used by the tobacco industry to deny direct causation).
- **Collider:** Talent $\rightarrow$ Celebrity *and* Luck $\rightarrow$ Celebrity.

Conditioning: The "Valve" Intuition

How does information flow when we "control" for a variable?

- **Chains & Forks:** These paths are **open** by default. Conditioning on the middle variable (M or Z) **blocks** the flow.
- **Colliders:** These paths are **blocked** by default. Conditioning on the collider (C) **opens** the path.

The Golden Rule
Conditioning on a common **cause** removes bias. Conditioning on a common **effect** creates bias.

Bioscience Examples

- **Gene Regulation:** Gene A activates Gene B, which suppresses Gene C ($A \to B \to C$).
- **Pleiotropy (Fork):** A single genetic variant affects both height and bone density.
- **Selection Bias (Collider):** In a study of hospitalized patients, we might find a fake link between two unrelated diseases because "being in the hospital" is a common effect of both.

Three Routes to Causal Knowledge

How do we find the "arrows"?

1. **Experiments (RCTs):** Physically manipulate the system to break confounding.
2. **Quasi-Experiments:** Use design-based "as-if random variation (e.g., IV, Diff-in-Diff, Regression Discontinuity) to mimic experiments.
3. **Causal Discovery:** Use algorithms to infer the arrows from purely observational data (Sucar 2026 focus).

RCT: The Reference Design

- **Why randomization helps:** by randomly assigning X, treatment assignment is (approximately) independent of pre-treatment causes (confounders).
- **Graph view:** randomization **breaks incoming arrows into** X (in expectation), so comparisons approximate $do(X)$ rather than selection.
- **What RCTs typically identify:** an **intention-to-treat** effect; per-protocol effects require extra assumptions/analysis.

Practical caveats (still matter in RCTs)
Noncompliance, attrition/missingness, interference/spillovers, measurement error, and finite-sample imbalance can bias or weaken conclusions. External validity (generalization) is separate from identification.

Quasi-experiments: When you cannot randomize

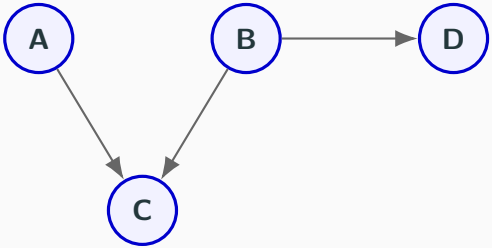
Used when ethics or logistics prevent an RCT (e.g., we can't force people to smoke).

- **Instrumental Variables (IV):** Finding a variable that affects X but has no direct link to Y.
- **Difference-in-Differences:** Comparing changes over time between a treated group and a control group.
- **Regression Discontinuity:** Comparing people just above and just below a strict cutoff (e.g., a test score).

Causal Discovery: Learning Structure from Data

- **Core Idea:** If we have enough variables, the patterns of *conditional independence* in the data act like "fingerprints" of the underlying DAG.
- **Faithfulness Assumption:** We assume that the independencies we see in the data are due to the graph structure, not a weird coincidence of numbers.
- **The Output:** Often a **CPDAG** (Completed Partially Directed Acyclic Graph) where some arrows are directed and others remain ambiguous (or a **PAG** if latent confounding is allowed).

Exercise: Read the DAG



1. Are A and B independent?
2. If we condition on C, do A and B become dependent?
3. Identify the **Collider** in this graph.

Session 2: Causal Inference – Identification and Reasoning

Learning Outcomes

By the end of this session, you will be able to:

- **Distinguish** between $P(Y|X)$ (seeing) and $P(Y|do(X))$ (doing).
- **Select** valid adjustment sets using the Backdoor Criterion.
- **Recognize** "Bad Controls" (Mediators and Colliders).
- **Conceptualize** Counterfactuals via the Abduction-Action-Prediction workflow.

What is Causal Inference?

- It is the study of how one variable X **influences** another variable Y within a system.
- **Moving beyond probability:** Traditional statistics deals with $P(Y \mid X)$. Causal inference deals with $P(Y \mid do(X))$.
- **The SCM approach (Sucar 2026):** We assume a set of Structural Equations where $Y = f(X, U)$. This represents the *mechanism* rather than just a correlation.

Causal Inference Workflow

According to Sucar (2026), the transition from data to causal claims follows this path:

1. **Modeling:** Draw the DAG (or discover it from data) based on domain knowledge.
2. **Identification:** Use the DAG and the "Calculus of Intervention" to see if $P(Y \mid do(X))$ (the causal query) can be calculated from observed data.
3. **Estimation:** Use statistical methods (regression, matching, IPW) to calculate the values.
4. **Robustness:** Test if the result holds under different assumptions – if it survives "stress tests" (e.g., sensitivity to hidden confounders).

Prediction vs Intervention

Prediction (Passive)

- "Given that I **see** X, what is Y?"
- Does not require a graph.
- Fails when the environment changes or if the system is disturbed.

Intervention (Active)

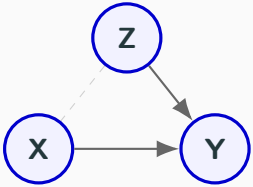
- "If I **change** X to x, what is Y?"
- Requires a causal model (DAG).
- Stable under policy changes.

The Difference
Conditioning is a "filter" on existing data. Intervention is a "mutilation" of the physical process.

Classic Example
Predicting rain by seeing people with umbrellas (Prediction) vs. forcing people to open umbrellas to try and make it rain (Intervention).

Conditioning vs Intervention

- **Conditioning ($X = x$):** Selection bias.
  We focus on a sub-population where X naturally occurred and is still influenced by its parents (Z).
- **Intervention ($do(X = x)$):** Structural change.
  We "mutilate" the graph by deleting all arrows pointing *into* X.
  - X is a constant and is no longer a function of its parents.
- $P(Y \mid X = x)$ is usually **not** equal to $P(Y \mid do(X = x))$.



Mutilated Graph $G_{\bar{x}}$

Backdoors: The Source of Confounding

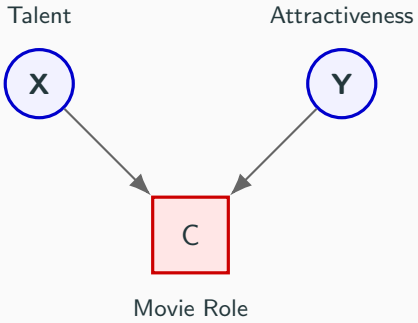
- A **backdoor path** is any non-causal path between X and Y that begins with an arrow into X ($X \leftarrow \cdots \rightarrow Y$).
- These paths "leak" association, making X and Y look related when they aren't.
- **The Solution:** Adjust for a set Z that "blocks" these paths.

$$P(Y \mid do(X)) = \sum_z P(Y \mid X, Z = z) P(Z = z)$$

For the **total effect** *via backdoor adjustment, avoid adjusting for* **descendants of** X *(post-treatment variables). Adjusting for mediators/colliders changes the question or can introduce bias.*

Colliders: The Selection Trap

- A **Collider** is a variable C where two arrows "collide" ($X \to C \leftarrow Y$).
- **The Trap:** Conditioning on C creates a spurious correlation between X and Y, opening a path between X and Y.
- **Selection Bias:** Common in hospital or recruitment data (e.g., Talent vs. Attractiveness in Hollywood).



Mediators: The Mechanism Path

- **Mediator:** M carries part of the causal effect ($X \to M \to Y$).
- **Total effect (common target):** do **not** condition on M if you want the overall impact of changing X on Y.
- **Direct-effect-type question (conceptual):** conditioning on M *changes the estimand* by blocking the mediated pathway.
- **Caution:** formal direct/indirect decomposition can require additional assumptions; naive conditioning on M is not automatically a valid causal direct-effect estimate.

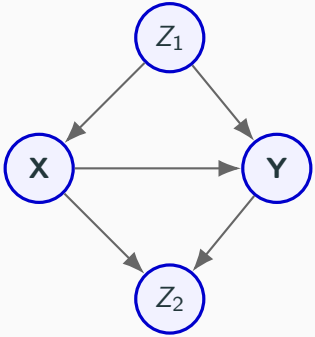
Smoking (Tar) Cancer

Conditioning blocks $X \to M \to Y$

Counterfactuals

- Counterfactuals answer: "What would have happened if . . . ?" (for the same unit/context).
- **Sucar's 3-Step Algorithm:**
  1. **Abduction:** update beliefs about unobserved noise U given evidence e.
  2. **Action:** modify the model M to M_x by $do(X = x)$; i.e., intervene on the SCM.
  3. **Prediction:** compute the counterfactual outcome Y using the updated U under the modified model.
- This supports **unit-level** "same unit, different action" reasoning. In practice, counterfactual and do-operator views are complementary formalisms for causal queries.

Exercise: Valid Adjustment Sets



1. Is there a backdoor path? If so, identify it.
2. Is there a collider? If so, identify it.
3. What is the valid adjustment set to find the **Total Effect** of X on Y?
4. Which variable(s) must you *avoid* adjusting for?

Session 3: Causal Discovery – The Cartography of Biological Mechanisms

Learning Outcomes

By the end of this session, you will be able to:

- **Explain** the theoretical limits of learning causality from purely observational bio-data.
- **Identify** the Markov Equivalence Class (MEC) of a biological signaling pathway.
- **Navigate** the Discovery Master Map: choosing algorithms based on data regime (Obs/Int/Temp).
- **Apply** a reporting discipline that distinguishes between high-confidence "stable" edges and ambiguous correlations.

Discovery vs. Inference: Where are we?

Causal Inference (Session 2)

- **Input:** A known DAG (e.g., a literature-derived pathway).
- **Task:** Calculate the effect size of a drug or mutation.
- *"If I knock out Gene X, what happens to Phenotype Y?"*

Causal Discovery (Sessions 3–6)

- **Input:** Data (RNA-seq, Proteomics, CRISPR screens).
- **Task:** Learn the arrows (the structure).
- *"Which genes regulate this metabolic pathway?"*

The AI for Bioscience Challenge
In many NILAB projects, the "map" is incomplete or unknown. Discovery is the process of **Automated Hypothesis Generation**.

The Observational Wall: Markov Equivalence

Why can't we always find the "true" biological graph from steady-state data?

- **Observational Equivalence:** Multiple biological mechanisms can produce the exact same correlation patterns.
- **The V-Structure Exception:** $G_1 \rightarrow G_2 \leftarrow G_3$ (Two genes regulating one target) is unique and identifiable.
- **The CPDAG:** Algorithms usually return a **Completed Partially Directed Acyclic Graph**—a mix of directed ($\rightarrow$) and undirected ($-$) edges.

Equivalence Class (MEC)

Represents three possibilities:

$G1 \rightarrow G2 \rightarrow G3$

$G1 \leftarrow G2 \leftarrow G3$

$G1 \leftarrow G2 \rightarrow G3$ (Fork)

The Discovery Master Map: Algorithmic Families

Family	Logic	Bioscience Use-Case
Constraint-based (e.g., PC)	Elimination via Conditional Independence tests.	Large-scale multi-omics (transcriptomics + clinical).
Score-based (e.g., GES/BIC)	Global search for the graph that maximizes "Fit vs. Sparsity."	Metabolic flux analysis or protein signaling pathways.
Functional / stronger-assumption (e.g., LiNGAM, ANM)	Gains orientation by adding assumptions (e.g., linear non-Gaussian noise; functional/noise asymmetry).	Small-to-moderate gene sets where assumptions are plausible; use as hypothesis generation with explicit checks.

The Discovery Master Map: Data Regimes

- **Observational (Session 4):** Cross-sectional data. Identifies the MEC.
- **Interventional (Session 5):** CRISPR knockouts, drug perturbations. Breaks equivalence; orients arrows.
- **Temporal (Session 6):** Time-course metabolomics. Time-order provides a "causal anchor."

Sucar's Insight (2026)
Effective AI discovery in bioscience often requires **Hybrid Discovery**: using a constraint-based skeleton with a score-based orientation phase.

Reporting Discipline: "Hypothesis Generation"

Causal Discovery is rarely about "The True Graph"; it is about **Robust Hypotheses**.

1. **Assumptions Disclosure:** Are we assuming *Causal Sufficiency* (no unmeasured regulators)?
2. **Stability Analysis:** If we remove 10% of the cells/samples, does the edge $GeneA \rightarrow GeneB$ vanish?
3. **Object Labeling:** Did you find a **DAG** (unlikely), a **CPDAG** (common), or a **PAG** (includes latent drivers)?
4. **Wording:** "The edge is consistent with the data" vs. "We proved causation."

Activity: Scenario Selection

Scenario: You have RNA-seq data for 2,000 genes from 500 patients. You also have limited CRISPR knockout data for the 10 most "central" genes.

Group Task:

1. Which **Data Regime** applies? (Obs, Int, or Hybrid?)
2. Which **Algorithmic Family** would you start with?
3. What is the biggest **Risk** to your discovery? (e.g., Latent confounding, small n, non-linearity).
4. How would you report a link between a gene and a disease if it only appears in the interventional data?

Session 4: Observational Discovery – Learning Biological Networks from Steady-State Data

Learning Outcomes

By the end of this session, you will be able to:

- **Execute** the logic of the PC Algorithm (Skeleton $\rightarrow$ V-structures $\rightarrow$ Meek Rules).
- **Evaluate** the trade-off between Likelihood and Sparsity using the BIC score.
- **Select** appropriate p-value (α) and penalty thresholds for high-dimensional bio-data.
- **Identify** when a "Hidden Driver" (Latent Confounder) is likely corrupting your observational results.

The Observational Setting: Identifiability Limits

In the NILAB context, this is often "Snapshot" data: n samples of d variables (e.g., 500 tumor biopsies).

- **Identifiability:** We can learn the **adjacencies** (who interacts with whom).
- **The Orientation Problem:** We can only orient edges that participate in a **V-structure** ($G1 \rightarrow G3 \leftarrow G2$) or those forced by the "no-cycle" rule.
- **The Result:** A CPDAG. If an edge is undirected ($G1 - G2$), the data cannot tell you the direction.

Bio-Insight
If you see an undirected edge between a transcription factor and a gene, observational data alone cannot confirm which is the regulator.

Constraint-based: The PC Algorithm Loop

The PC algorithm (named after Spirtes and Glymour) is the most common constraint-based method.

1. **Skeleton Phase:** Start with a fully connected graph. Perform Conditional Independence (CI) tests. If $G_i \perp G_j \mid \mathbf{Z}$, remove the edge.
2. **V-structure Phase:** For all $G_i - G_j - G_k$, check if the "middle" node G_j was in the set that separated G_i and G_k. If not, orient as $G_i \rightarrow G_j \leftarrow G_k$.
3. **Meek Rules:** Orient remaining edges to avoid creating new V-structures or cycles.

Score-based: Balancing Fit and Parsimony

Score-based methods (like **GES**) view discovery as an optimization problem: search for the graph G that maximizes the score.

$$BIC(G, D) = \ln P(D \mid \hat{\theta}, G) - \frac{k}{2} \ln n$$

- **Likelihood ($\ln P$):** How well does the model explain the biological correlations?
- **Penalty (k):** Penalizes complexity (number of arrows).

The Parsimony Principle

In Bioscience, we assume biology is "sparse." We prefer a model with 5 high-confidence arrows over one with 50 weak ones.

Worked example: steady-state network discovery (observational)

- **Setting:** $n = 500$ samples, $d = 2{,}000$ genes (RNA-seq). Goal: discover a *local* regulatory network around a pathway (e.g., 30–80 genes).
- **Step 1 (scope):** reduce d using biology + statistics (pathway databases, differential expression, highly variable genes). Discovery is unreliable on an unfiltered $d \gg n$ "hairball".
- **Step 2 (skeleton):** run a constraint-based method (PC) with an appropriate CI test (e.g., partial correlations for approximately Gaussian data; nonparametric CI tests otherwise). Tune α and limit conditioning set size.
- **Step 3 (orientation):** interpret the output as a **CPDAG** (not a fully identified DAG). Undirected edges are **genuinely ambiguous** from observational data.
- **Step 4 (robustness):** bootstrap / subsample stability (e.g., report edges appearing in > 70–80% of runs). Sweep α (PC) or penalties (GES/BIC).
- **Step 5 (validation plan):** choose 1–2 edges for follow-up perturbation (Session 5), prioritizing stable edges that affect a phenotype readout.

Teaching message

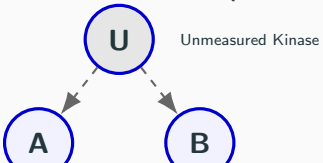
Observational discovery yields **hypotheses + uncertainty**. The professional output is a stable subset + a clear plan for validation.

Why Discovery Fails: The "NILAB" Warnings

Observational discovery relies on the **Faithfulness** and **Sufficiency** assumptions. In biology, these are often violated:

- **Latent Confounders:** You measure Protein A and B, but not the Kinase U. The algorithm wrongly detects a direct link because $A \not\perp B$.
- **Selection Bias:** Samples only from "surviving cells" creates a collider effect, inducing a fake correlation between independent pathways.
- **Measurement Noise:** Technical noise in scRNA-seq can mask independence, leading to "hairball" graphs (over-connected).

True Mechanism (Reality)



$\Longrightarrow$

Algorithmic Output (Error)

Exercise: Interpretation of Discovery Outputs

You run two algorithms on a dataset of 10 proteins.

Algorithm A (PC, $\alpha = 0.01$)

- Result: 5 directed edges, 12 undirected edges.
- Sparse graph.

Algorithm B (GES, low penalty)

- Result: 25 directed edges.
- Dense "hairball."

Discussion Questions:

1. Why does Algorithm A have undirected edges while B has directed ones?
2. Which result is more useful for designing a follow-up knockout experiment?
3. How would you validate the "extra" edges in Algorithm B?

Session 5: Interventional Discovery – Perturbations and the Operational Meaning of Causality

Learning Outcomes

By the end of this session, you will be able to:

- **Explain** how perturbations (CRISPR, drug screens) resolve Markov Equivalence.
- **Describe** the graph-theoretic effect of a "perfect" vs. "soft" intervention.
- **Select** optimal intervention targets to maximize "Information Gain."
- **Discuss** the trade-offs in Joint Discovery (GIES) when pooling observational and experimental bio-data.

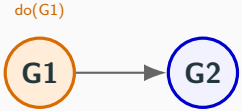
Breaking the Observational Wall

In Session 4, we saw that steady-state data leaves us with undirected edges (CPDAGs).

- **Key gain from perturbations:** interventions provide *extra information* that shrinks the Markov Equivalence Class (MEC) into an **interventional** equivalence class.
- **What a response means:** if intervening on A changes the distribution of B, then A is a **causal ancestor** of B (possibly via intermediates).
- **What no response means:** lack of change does **not** imply $B \to A$; it may indicate no causal effect, compensation, insufficient power, or an imperfect intervention.
- **Bio-reality:** in NILAB projects, this is the transition from "association omics" to perturbation data (CRISPR, Perturb-seq, drug screens).

Breaking the Observational Wall (cont)

Observational: direction ambiguous



Interventional: ambiguity reduced

Intervention Types in Bioscience

According to Sucar (2026), how we model the intervention depends on the biological protocol:

Hard (Atomic) Interventions

- **Example:** Gene Knockout (CRISPR).
- **Graph Change:** "Mutilates" the graph by deleting all incoming arrows into the target.
- $G1$ is no longer controlled by its biological regulators.

Soft (Shift) Interventions

- **Example:** Drug titration or overexpression.
- **Graph Change:** Does not break incoming arrows; it just changes the conditional distribution $P(G1 \mid Parents)$.
- The biological "regulators" are still active but "shuffled."

Joint Discovery: Learning from Everything

We rarely have *only* interventional data. **GIES (Greedy Interventional Equivalence Search)** pools all data types.

- **Input:** A collection of datasets (e.g., 1,000 Wild-Type cells + 100 Knockout cells per gene).
- **The Search:** Like GES, it optimizes a score (BIC), but the score is calculated across different "mutilated" versions of the graph for each intervention batch.
- **Benefit:** High sensitivity. Even a few knocked-out samples can orient a whole pathway.

Worked example: hybrid discovery with CRISPR/drug perturbations

- **Setting:** baseline (wild-type) data + k perturbation conditions (e.g., CRISPR knockout of 10 candidate regulators; 100–500 cells per condition).
- **Start point:** learn an observational CPDAG (Session 4) to propose candidates and identify ambiguous edges/clusters.
- **Perturbation logic:** if $do(A)$ changes B, conclude A is a **causal ancestor** of B (not necessarily a direct edge). Lack of change is inconclusive without power/protocol checks.
- **Joint modeling:** pool observational + interventional data (e.g., GIES/GIES-like scoring) to reduce equivalence and orient edges.
- **Bio pitfalls to control:** batch effects across plates/days, incomplete knockouts, off-target effects, and compensatory pathways.
- **Reporting:** separate (i) stable oriented edges, (ii) ambiguous edges remaining, and (iii) protocol details enabling reproducibility.

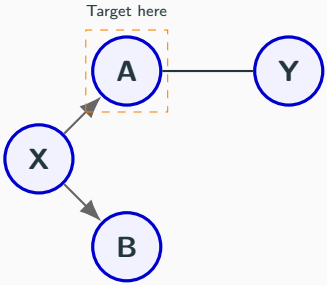
Teaching message

Interventional data improves identifiability, but interpretation must separate **direct edges** from **downstream responses** and must report protocol caveats.

Intervention Planning: Optimal Target Selection

Interventions are expensive (reagents, time, compute). Where should we perturb?

- **Hub Nodes:** Perturbing a highly connected "master regulator" provides global information but can be noisy.
- **Ambiguous Pairs:** Target nodes at the ends of undirected edges ($A - B$) to force orientation.
- **Downstream Effects:** Intervene "upstream" to see how signals propagate through the cascade.



Bio-Caveats: When $do(X)$ isn't just $do(X)$

Causal Discovery algorithms assume "Perfect Targetting." Bio-AI must account for:

- **Off-Target Effects:** CRISPR might accidentally hit a homologous gene. The "mutilation" is wider than intended.
- **Batch Effects:** Interventional data often comes from different plates/days than observational data.
- **Incomplete Penetrance:** Not every cell in the "knockout" group actually has a successful knockout.

The Discovery Rule
If the data shows a change in a gene that is *not* downstream of your target, you may have an off-target effect or a hidden confounder.

Activity: The NILAB Intervention Challenge

Scenario: You have a CPDAG of a signaling pathway with 15 proteins. You have 3 undirected clusters (undirected edges). You have the budget for **two** CRISPR knockouts.

Your Task:

1. Look at the provided graph (Handout). Which two proteins provide the most "structural information"?
2. Would you prefer a "Hard" knockout or a "Soft" drug inhibition to find the direction?
3. If you intervene on $P1$ and $P5$ changes, what can you conclude about the edge $P1 - P5$?

Session 6: Temporal Discovery – Lagged Effects and Dynamic Networks

Learning Outcomes

By the end of this session, you will be able to:

- **Differentiate** between Granger Causality and Structural Temporal Discovery.
- **Construct** a 2-Time-slice Bayesian Network (2-TBN) template.
- **Diagnose** sampling rate pitfalls (Aliasing) in biological time-series.
- **Design** a validation pipeline for non-stationary biological processes.

Time as a Causal Anchor

Temporal ordering helps, but it does not remove confounding.

- **Ordering constraint:** causal influence can only go from past to future, so lagged edges are restricted to $t-1 \to t$ (and earlier).
- **Predictive precedence is not enough:** $G1_{t-1}$ predicting $G2_t$ suggests a candidate influence, but structural interpretation still requires controlling for autocorrelation and common drivers.
- **Autocorrelation (memory):** always control for self-lags (e.g., $G1_{t-1} \to G1_t$) before claiming cross-lag effects.
- **Granger vs structural discovery:** Granger tests predictive improvement; DBN-style discovery aims to learn a structural (assumption-dependent) mechanism graph over time.

Teaching message

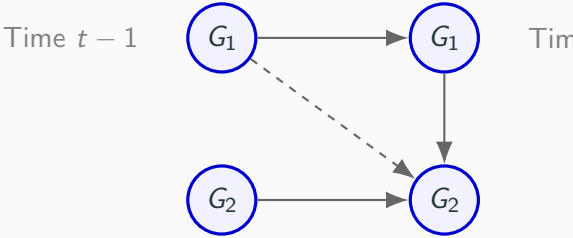
Time provides an **ordering constraint** (past→future), not a guarantee of causality; robustness checks remain essential.

The 2-TBN Framework

Sucar (2026) represents temporal causality using the **2-Time-slice Bayesian Network (2-TBN)**.

- **Intra-slice edges:** Instantaneous interactions (within the same measurement).
- **Inter-slice edges:** Delayed interactions (from $t-1$ to t).
- **Stationarity:** We assume the "rules" of the network don't change as the experiment progresses.

Unrolled Graph



Worked example: time-course metabolomics (2-TBN/DBN reasoning)

- **Setting:** 48-hour time-course of 50 metabolites after insulin; sampling every hour. Goal: lagged influence structure (who drives whom over time).
- **Model choice:** represent dynamics with a **2-time-slice template** (lags from $t-1$ to t) and include **self-lags** for each variable.
- **Key knob:** choose a maximum lag $\tau_{\max}$ (e.g., 1–6 hours). Report sensitivity: do key edges persist when $\tau_{\max}$ changes?
- **Confounding control:** control for autocorrelation and shared drivers (e.g., circadian effects, media changes). "Predicts" does not automatically mean "causes".
- **Nonstationarity check:** run discovery in sliding windows (early vs late) to see whether mechanisms change after insulin response peaks.
- **Interpretation policy:** prioritize stable lagged edges; treat contemporaneous edges as ambiguous without higher-frequency measurement or additional assumptions.

Teaching message

Time helps ordering, but **lag choice + stationarity + hidden drivers** determine whether a temporal graph is a useful causal hypothesis.

The Sampling Pitfall: When Biology is Faster than AI

In NILAB, we often measure protein levels every hour. But what if the signaling happens in seconds?

- **Aliasing:** A causal chain $A \to B \to C$ looks like a simultaneous "hairball" if the sampling interval is too wide.
- **Hidden Feedback:** If you miss the intermediate steps, a circular feedback loop might look like a bi-directed (confounded) edge.
- **Solution:** Sensitivity analysis across different lags (τ). Does the structure disappear if you skip every second time-point?

Bio-Sampling rule-of-thumb (Nyquist-inspired)
Sample faster than the fastest causal dynamics you aim to resolve; otherwise intermediate steps are aliased and edges can be mis-attributed.

Non-Stationarity: Changing Mechanisms

Does $G1 \rightarrow G2$ always hold?

- **Differentiation:** The regulatory network of a stem cell changes as it becomes a neuron.
- **Stress Response:** A drug treatment may trigger a "regime shift" where new causal arrows appear.
- **Detection:** Use a "Sliding Window" discovery. Run the algorithm on the first 10 time-points, then the next 10. Do the arrows match?

Capstone: A Responsible Discovery Pipeline

To publish "Discovery AI" results in a BBSRC-level journal, follow this policy:

1. **Preprocess:** Handle missing data and normalize (log-transform) bio-signals.
2. **Discovery:** Use a Hybrid approach (Constraint skeleton + Score orientation).
3. **Stability:** Run Bootstrapping. Only report edges that appear in $> 80\%$ of runs.
4. **Validation:** Compare discovered arrows against databases (StringDB, KEGG) and propose **one** specific validation experiment (Session 5).

Discovery is the start of the scientific method, not the end.

Final Activity: Temporal Omics Pipeline

Data: A 48-hour time-course of 50 metabolites after a dose of Insulin.

Tasks:

1. Draw the 2-TBN template for three nodes: *Insulin*, *Glucose*, *Pyruvate*.
2. If $Glucose_t$ and $Pyruvate_t$ are highly correlated, but no lagged edge exists, what does this suggest?
3. How would you check if the "Glucose-regulation" mechanism is stationary over the 48 hours?